

Hassen Slim

Software Engineer | Applied AI | Backend

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Summary

Software Engineer currently pursuing a Master 2 at Université Grenoble Alpes, with experience across **backend architecture**, **distributed systems**, **large-scale data processing**, **machine learning**, **deep learning**, **computer vision**, **agentic AI** and **geospatial systems**. Interested in designing **reliable software** and **applied AI systems** for **real-world scientific**, **industrial** and **data-driven applications**.

Education

Université Grenoble Alpes (UGA)

Grenoble, France

Master 2 Ingénierie de la Santé – parcours MITI

Sep 2026 – Present

Models, innovation, technologies, data and computational methods applied to scientific and health-related systems.

École Nationale d'Ingénieurs de Tunis (ENIT)

Tunis, Tunisia

National Engineering Degree in Software Engineering

Sep 2023 – Sep 2026

Selected coursework: Software Architecture, Distributed Systems, Cloud Computing, Machine Learning, Deep Learning, NLP & Generative AI, Semantic Web, Cybersecurity, Secure Web Applications, Software-Defined Networking, IoT Middleware and Big Data.

Preparatory Classes – IPEIN / IPEIM

Tunisia

Ranked **379 / 1830** at the national competitive entrance examination for engineering schools.

Sep 2021 – Jun 2023

Professional & Research Experience

IMT Atlantique – Lab-STICC (in collaboration with SHOM)

Brest, France

Software Engineering & AI Research Intern

Mar 2026 – Sep 2026

- Refactored a large scientific and geospatial Python backend toward **Clean Architecture** and **CQRS**, separating FastAPI routing, application handlers, domain logic, repositories and persistence layers.
- Developed **REST** and **OGC WFS 2.0** services using **FastAPI**, **Pydantic v2**, **async SQLAlchemy 2.0** and **PostgreSQL/PostGIS**, including authentication, authorization, domain validation and transaction management.
- Implemented WFS operations including **GetCapabilities**, **DescribeFeatureType** and **GetFeature** for interoperable geospatial data access.
- Optimized ingestion of approximately **500,000 geographic features**, reducing execution time from approximately **625 seconds** to **70 seconds** through PostgreSQL **COPY**, vectorized geometry encoding, Pyogrio-based reading and lower-overhead processing.
- Investigated asynchronous runtime and database lifecycle issues involving workers, event loops and connection pools to improve reliability under long-running processing jobs.
- Developed a two-stage machine-learning pipeline combining **multi-output XGBoost** for missing-variable reconstruction with a **Graph Attention Network (GAT)** for spatial refinement over graph-structured maritime zones.
- Designed controlled masking experiments to evaluate reconstruction quality while preserving observed scientific measurements as ground truth.

WE DIGITAL Solutions

Tunisia

Software Engineer – Part-time

Oct 2025 – Jan 2026

- Contributed to a distributed e-commerce backend based on **Spring Boot microservices** and asynchronous communication with **Apache Kafka**.
- Worked with choreography-based service interactions, the **database-per-service** pattern and containerized deployment.
- Used **Docker**, **Kubernetes** and **GitLab CI** for application packaging, integration and deployment workflows.

Insomea Cloud Solutions

Tunisia

AI Engineering & DevOps Intern

Jun 2025 – Aug 2025

- Developed an AI-assisted backend integrating **LangChain**, **Google Gemini** and **Model Context Protocol (MCP)** for multi-step cloud and CI/CD automation.
- Implemented structured tool invocation and interaction between LLM reasoning components and infrastructure operations.
- Containerized and orchestrated services using **Docker** and **Kubernetes on Azure Kubernetes Service (AKS)**.
- Worked with cloud-native deployment tooling including **Helm**, **ArgoCD** and **CI/CD pipelines**.
- Analysed operational failure cases involving tool errors, missing parameters and unreliable LLM outputs.

Artificial Intelligence, Machine Learning & Data Projects

Graph-Based Reconstruction of Maritime Environmental Data

2026

- Developed a two-stage predictive architecture combining **multi-output XGBoost** and **Graph Attention Networks** for incomplete and spatially structured environmental data.

- Evaluated predictions through controlled feature masking, quantitative error analysis and comparison with known observations.

LangGraph Agentic RAG System

2025 – 2026

- Built an agentic document-question-answering backend using **FastAPI**, **LangGraph**, **Retrieval-Augmented Generation (RAG)**, **FAISS vector search** and **LLM reasoning**.
- Implemented stateful multi-step workflows including retrieval, memory, iterative reasoning, validation and human feedback.

MCP Server / LangGraph Client – Structured Tool Orchestration

2025

- Designed a Python **Model Context Protocol server** and LangGraph-based client for structured tool invocation and modular interaction between AI components.

Lung Cancer Detection from Histopathology Images

2025

- Built a biomedical image-classification pipeline using **transfer learning with VGG16 and ResNet50**.
- Implemented image preprocessing, model training, fine-tuning, validation and quantitative performance evaluation on histopathology images.
- Studied transfer learning, generalization and overfitting behaviour for medical image classification.

Multimodal Medical Image Analysis with PyTorch

2025

- Prototyped a **multimodal PyTorch architecture** combining visual representations from a **Vision Transformer (ViT)** with contextual information.
- Implemented **cross-attention** between visual and contextual representations and compared the multimodal architecture with an image-only baseline.

Farm Management AI Platform

Jun 2025 – Aug 2025

- Developed a full-stack platform using **Spring Boot**, **React** and **MongoDB** for farm monitoring and decision support.
- Integrated two **YOLOv8 computer-vision models** for crop and livestock anomaly detection.
- Connected image preprocessing and model inference services with backend application logic and user-facing workflows.

Automatic License Plate Detection and Recognition

Academic Project

- Developed a Python computer-vision pipeline using **YOLOv8**, **OpenCV** and **Tesseract OCR** for license-plate localization and text extraction.
- Integrated detection and OCR processing with structured application data handling.

Churn Prediction with Tree-Based Machine Learning

2025

- Compared **Random Forest** and **XGBoost** for prediction from structured tabular data.
- Performed preprocessing, feature analysis, training and model evaluation.

Software, Data Engineering & Distributed Systems Projects

Big Data Sales Analytics Pipeline – Hadoop MapReduce

Nov 2025

- Built a containerized **Hadoop cluster** and implemented distributed **Java MapReduce** jobs for large-scale retail-data analysis.
- Processed transaction data to extract statistics related to payment behaviour, sales and store performance.

University Courses Ontology – Semantic Web & Knowledge Graph

2025 – 2026

- Designed an academic knowledge model using **OWL/RDF** to represent courses, prerequisites, professors, departments and students.
- Implemented **property chains**, **SWRL rules** and **SPARQL queries** for automatic prerequisite inference, course eligibility and faculty workload analysis.
- Used ontology reasoning to verify logical consistency and infer relationships not explicitly stored in the knowledge base.

Distributed Microservices Architecture

2025

- Designed containerized backend services using **FastAPI**, **PostgreSQL**, **Docker** and **Kubernetes**.
- Explored REST-based inter-service communication, service isolation and independent persistence.

Java Document Search Engine

Academic Project

- Developed a Java search engine supporting **document indexing** and **keyword-based information retrieval**.
- Implemented search across multiple document formats with ranked results and contextual document snippets.

Cinema Reservation Web Application – Java/Jakarta EE

Academic Project

- Developed a Java enterprise web application for cinema reservation workflows using **Jakarta EE / JSP / Servlets**.
- Applied layered web application concepts, server-side processing and persistent application data management.

Administrative Dashboard – Angular

Frontend Project

- Developed an administrative web dashboard using **Angular** and **TypeScript**, including reusable UI components and frontend application structure.

Additional Academic & Engineering Work

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- **Big Data:** implemented data-processing exercises and pipelines using **Hadoop MapReduce**, **Hive**, **Apache Spark** and **Kafka**.

- **Remote Sensing & Geospatial Analysis:** worked with spatial and satellite data using **QGIS, ENVI, GeoPandas, OSMnx, Cartopy, Sentinel and Landsat**, including experimentation with **U-Net and DeepLabV3+** for image segmentation.
- **Software-Defined Networking:** worked with **OpenFlow, Open vSwitch and Mininet** to configure and analyse programmable network behaviour.
- **Security:** studied secure web systems, authentication, authorization, TLS/SSL, network security and application security principles.
- **Cloud-Native Systems:** worked with Docker images, Kubernetes deployments, services, ingress, Helm charts, ArgoCD and automated CI/CD workflows.
- **Software Architecture:** applied Clean Architecture, CQRS, microservices, event-driven communication and distributed-system patterns.

Technical Skills

- **Programming & Software Engineering:** Python, Java, SQL, Clean Architecture, CQRS, Microservices, Distributed Systems
- **Backend & Data:** FastAPI, Spring Boot, REST APIs, SQLAlchemy, PostgreSQL/PostGIS, MongoDB, Redis
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, CNNs, Graph Attention Networks, Vision Transformers, YOLOv8, transfer learning
- **Generative & Agentic AI:** LangChain, LangGraph, RAG, LLMs, MCP, embeddings, FAISS, vector search
- **Cloud, DevOps & Data Engineering:** Docker, Kubernetes, Azure AKS, Helm, ArgoCD, GitHub Actions, GitLab CI, Hadoop, Spark, Kafka
- **Tools & Domains:** Git, Linux, GeoPandas, QGIS, geospatial data processing, computer vision, scientific data processing
- **Languages:** Arabic (Native), French (C1 – TCF), English (B2)

Certifications

- **Building RAG Agents with LLMs** – NVIDIA Deep Learning Institute
- **Building RAG Apps Using MongoDB** – MongoDB
- **Building Transformer-Based Natural Language Processing Applications** – NVIDIA
- **Fundamentals of Deep Learning** – NVIDIA
- **Introduction to Transformer-Based Natural Language Processing** – NVIDIA
- **Microsoft Certified: DevOps Fundamentals** – Microsoft
- **CCNA: Introduction to Networks** – Cisco
- **Git and GitHub Mastery** – 365 Data Science